(1) If k is any constant, then $\int kf(x)dx = k\int f(x)\,dx$

(2) $\int (f_1(x) \pm f_2(x))\,dx = \int f_1(x)dx \pm \int f_2(x)dx$

If $\int f(x)\,dx = g(x)+c$, then $\int f(ax+b)\,dx = \frac{1}{a}\,g(ax+b)+c$

(1)	$\int \tan x\,dx = \log\lvert\sec x\rvert + c$
(2)	$\int \cot x\,dx = \log\lvert\sin x\rvert + c$
(3)	$\int cosecx\,dx = \log\lvert\text{cosec}\,x - \cot x\rvert + c$
(4)	$\int \sec x\,dx = \log\lvert\sec x + \tan x\rvert + c$

Bernoulli's formula for integration by Parts:

If u and v are functions of x, then the Bernoulli's rule is

$\int udv = uv - u'v_1 + u''v_2 + \ldots$

where u', u'', u''',... are successive derivatives of u and

v, v_1, v_2, v_2,..., are successive integrals of dv

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\sin bx - b\cos bx] + c$$

$$\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}[a\cos bx + b\sin bx] + c$$

$\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left\lvert\frac{a+x}{a-x}\right\rvert + c$	$\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + c$
$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left\lvert\frac{x-a}{x+a}\right\rvert + c$	$\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + c$
$\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left\lvert x+\sqrt{x^2-a^2}\right\rvert + c$	$\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left\lvert x+\sqrt{x^2+a^2}\right\rvert + c$

$$\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right) + c$$

$$\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left\lvert x+\sqrt{x^2-a^2}\right\rvert + c$$

$$\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left\lvert x+\sqrt{x^2+a^2}\right\rvert + c$$

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